

COA DATABASE DESCRIPTION

Prepared for
U.S. Department of Energy
Bartlesville Project Office

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1.0 INTRODUCTION

The composition and physical properties of crude oil vary widely from one reservoir to another within an oil field, as well as from one field or region to another. Although all oils consist of hydrocarbons and their derivatives, the proportions of various types of compounds differ greatly. This makes some oils more suitable than others for specific refining processes and uses. To take advantage of this diversity, one needs access to information in a large database of crude oil analyses. The Crude Oil Analysis Database (COADB) currently satisfies this need by offering 9,056 crude oil analyses. Of these, 8,500 are United States domestic oils. Table 1–1 shows a summary of analyses by location.

Table 1–1 Summary of Analyses in COA Database by Location

Location	Number of Analyses	Location	Number of Analyses
Alabama	20	North Dakota	127
Alaska	49	Ohio	58
Arizona	4	Oklahoma	1298
Arkansas	196	Pennsylvania	148
California	454	South Dakota	5
Colorado	278	Tennessee	2
Florida	9	Texas	2492
Illinois	149	Utah	122
Indiana	21	Virginia	2
Kansas	598	Washington	3
Kentucky	49	West Virginia	49
Louisiana	673	Wyoming	910
Michigan	115	Africa	101
Mississippi	130	Asia	105
Missouri	6	Europe	107
Montana	168	North America (except U.S.)	88
Nebraska	89	South America	90
Nevada	10	West Indies	25
New Mexico	248	Ocenia	48
New York	18		

1.1 HISTORY

Crude oils have been analyzed by routine methods in Bartlesville since the 1920s, when the facility was part of the United States Department of the Interior, Bureau of Mines.¹ By 1965 more than 7,000 crude oils had been analyzed, creating one of the largest collections of such information in the world.² After the data were digitized in the late 1970s, the crude oil analysis database retrieval system (COASYS) was implemented in 1980.³ This custom software application placed the COA Data Bank on-line and made it available to the public.

1.2 UPDATE

Since 1980 there have been major advances in data management software, telecommunications software, and hardware. To take advantage of these improvements, the Department of Energy (DOE) has now updated this system using a bulletin board system and a commercial relational database management system. The combination of these two components will allow continual improvements in the operation as well as expansion of the on-line services.

The new implementation has retained basic features of the old COASYS with enhanced user-friendly features to allow operation without frequent reference to the user's guide. This new implementation contains on-line help to show the user how to display location codes, county or subdivision codes, region codes, and definitions of terms needed to build a search of the data bank. The bulletin board system will allow the user to communicate with the system administrator through messaging capabilities. Information on new features and data can be announced on the bulletin board immediately after development. Thus, callers will not have to wait for the user's guide to be republished.

1.3 ANALYSIS PROCEDURES

The database contains results of analysis of the general properties and chemical composition, as well as the field, formation, and geographic location of the crude oil sample. Figure 1-1 shows the procedures used to analyze crude oil. The gravity, sulfur content, nitrogen content, viscosity, color, and pour point have been determined for most samples. The oil is also distilled up to an equivalent temperature of 418.6°C (785.5°F), obtaining fractions at 25°C (45°F) intervals. In addition to volume percent of each distillation fraction, the gravity, refractive index, viscosity, and cloud point of each fraction are obtained where appropriate. From this data, calculations can be made to determine the asphalt content, straight-run gasoline, hydrocarbon composition, and other properties. Table 1-2 lists some common oil distillation categories that are frequently used for oil correlations.

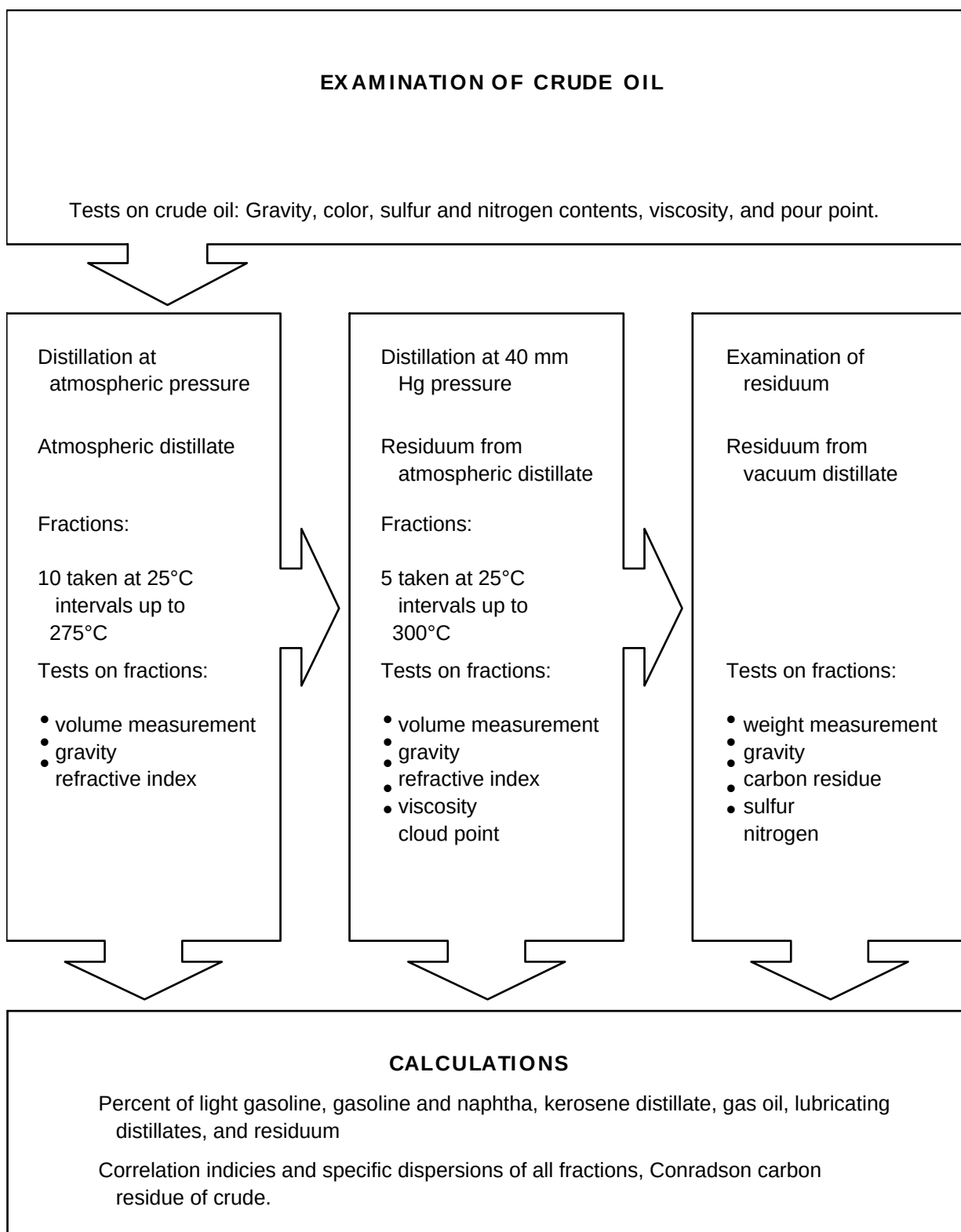


Figure 1-1 Procedure for Crude Oil Analysis

Table 1–2 Summary of Distillation Categories Used for Oil Correlations

Category	Description
Light Gasoline	Fractions up to 100°C (212°F).
Total Gasoline and Naptha	Fractions up to 200°C (392°F) except those with specific gravity greater than 0.825.
Kerosene	Fractions 200°–275°C (392°–527°F) having specific gravity below 0.825.
Gas Oil	Atmospheric distillation fractions above 0.825 specific gravity plus vacuum distillate with viscosity less than 50 SUS.
Nonviscous Lubricating Distillate	Vacuum distillate with viscosity between 50 and 100 SUS.
Medium Lubricating Distillate	Vacuum distillate with viscosity between 100 and 200 SUS.
Viscous Lubricating Distillate	Distillate with viscosity over 200 SUS.
Residuum	Remaining after distillation equivalent to 420°C (788°F).

1.4 NEW DATA

Supplemental information about crude oils currently in the data bank and information about new crude oils will be added to the data bank. Analyses of trace metals and hydrocarbon types that are of the most interest to industry—BETX (benzene, ethylbenzene, toluene, xylenes), and PNA (paraffins, naphthenes, aromatics)—have been started on oil samples that are in cold storage in Bartlesville. Additional data on the wells sampled will also be added in the future. The bulletin board screen will announce these additions and the search parameters you can use to retrieve this new data.

3.0 REFERENCES

1. Smith, N. A. C., H. M. Smith, O. C. Blade, and E. L. Garton. 1951. *The Bureau of Mines Routine Method for the Analysis of Crude Petroleum*, BuMines Bull. 490.
2. Smith, H. M. and J. H. Hale. 1966. *Crude Oil Characterizations Based on Bureau of Mines Routine Analyses*. BuMines, Report of Investigations 6846.
3. *BPO Crude Oil Analysis Data Bank User's Guide*. 1987. DOE Report No. DOE/BC-87/5/SP.
4. Clark Development Company, Inc. 1990. *PCBoard 14.5 User Reference Manual*.
5. Dudley, Marshall. 1978-1992. *DOORWAY Remote Access Program*, Version 2.2.
6. Microsoft Corporation, 1989-1993. *FoxPro Relational Database Management System for MS-DOS*, Version 2.5.

APPENDIX A

KEYWORDS AND DESCRIPTIONS

Keyword	Long Description
API	API gravity. Directly related to specific gravity at 60°F. As shown by this formula: $^{\circ}\text{API} = (141.5/\text{SPGR}) - 131.5$
CNTY	County or district code. For the United States, each state's counties are numerically coded. Foreign countries have numeric codes assigned for the various districts or provinces. See Appendix B for an index of these codes.
CRRES	Weight % carbon residue of residuum from vacuum distillation, obtained by Ramsbottom method and converted to Conradson equivalent
CRCO	Weight % carbon residue of crude—directly related to CRRES by the formula: $\text{CRCO} = (\text{RES}) (\text{SGRES}) (\text{CRRES}) / (\text{SGCO}) (100)$
DEPTH	Depth or first value of depth range
FIELD	Oil field name
FORM	Geological formation name
GEOCODE	Geological formation (AAPG mnemonic code, 4 characters)
SGGSP	Specific gravity at 60°F of total gasoline and naphtha fractions
SGGO	Specific gravity at 60°F of gas oil fraction
GSP	Total volume % gasoline and naphtha fraction
GO	Volume % gas oil

Keyword	Long Description
SGKD	Specific gravity at 60°F of kerosene distillate
KD	Volume % kerosene distillate
LGS	Volume % light gasoline
SGLGS	Specific gravity at 60°F of light gasoline distillate
LOC	Location Name Code. Each state in the U.S., each Canadian Province, and each foreign country for which the COA Database has sample data are given numeric codes. For a list of names and codes see Appendix B. (For offshore, search county code 900 and adjacent landward counties of appropriate state.)
MVLD	Volume % medium viscous lubricating distillate
N2CO	Weight % nitrogen in crude
N2RES	Weight % nitrogen in residue
NVLD	Volume % nonviscous lubricating distillate
PRPT	Pour point. Maximum temperature in °F at which a liquid will not pour
REGION	Geographical subdivisions within the states of Oklahoma or Texas
RES	Volume % residuum from vacuum distillation
SGRES	Specific gravity at 60°F of the residue
SGCO	Specific gravity. Ratio of the density of crude at 60°F to the density of water at 60°F
SRES	Weight % sulfur in residue
SCO	Weight % sulfur in crude

Keyword	Long Description
SU100	Viscosity (Saybolt Universal Seconds) at 100°F
SU77	Viscosity (Saybolt Universal Seconds) at 77°F
VLD	Volume % viscous lubricating distillate
YEAR	The keyword for year and sample identification code. The identification code is divided into three parts: • One letter for laboratory B Bartlesville L Laramie • Two digits for year For example 54 equals 1954 Samples prior to 1923 are numbered sequentially: 01, 02, etc. • Three digits for sequential numbering within a year 001–999 e.g., B75003: Bartlesville, 1975, Sample No. 3

APPENDIX B

Numerical Code for States and Counties of the United States

Index of Codes for Districts or Subdivisions of Foreign Countries

ALABAMA	STATE	COUNTY	Alabama (Cont.)	State	County
Autauga.....	1	1	Lowndes.....	1	85
Baldwin.....	1	3			
Barbour	1	5	Macon	1	87
Blount	1	7	Madison	1	89
Chilton	1	21	Marengo	1	91
			Marion.....	1	93
Choctaw	1	23	Marshall	1	95
Clarke.....	1	25			
Clay	1	27	Mobile.....	1	97
Cleburne.....	1	29	Monroe.....	1	99
Coffee	1	31	Montgomery.....	1	101
			Morgan.....	1	103
Colbert	1	33	Perry.....	1	105
Conecuh.....	1	35			
Coosa	1	37	Pickens	1	107
Covington	1	39	Pike	1	109
Crenshaw	1	41	Randolph.....	1	111
			Russell.....	1	113
Cullman.....	1	43	St. Clair	1	115
De Kalb.....	1	49			
Elmore.....	1	51	Shelby	1	117
Escambia.....	1	53	Sumter	1	119
Etowah	1	55	Talladega.....	1	121
			Tallapoosa	1	123
Fayette.....	1	57	Tuscaloosa	1	125
Franklin.....	1	59			
Geneva	1	61	Walker.....	1	127
Greene.....	1	63	Washington	1	129
Hale.....	1	65	Wilcox.....	1	131
			Winston.....	1	133
Henry	1	67			
Houston.....	1	69	ALASKA	STATE	COUNTY
Jackson.....	1	71	Aleutian Islands	50	5
Jefferson.....	1	73	Anchorage.....	50	10
Lamar	1	75	Barrow	50	15
			Bethel	50	20
Lauderdale	1	77	Bristol Bay	50	25
Lawrence.....	1	79	Cordova McCarthy.....	50	30
Lee	1	81	Fairbanks.....	50	35
Limestone	1	83	Juneau	50	40

Alaska (Cont.)	State	County
Kenai Cook Inlet.....	50	45
Kenai Peninsula	50	46
Ketchikan.....	50	50
Kobuk	50	55
Kodiak	50	60
Kuskokwim.....	50	65
Lynn Canal Icy	50	70
Nome.....	50	75
North Slope.....	50	77
Palmer Wasilla Talke.....	50	80
Prince of Wales.....	50	85
Seward	50	90
Sitka	50	95
Upper Yukon	50	100
Valdez Chitina Whitt.....	50	105
Wade Hampton	50	110
Wrangell Petersburg.....	50	115
Yukon Koyukuk	50	120

ARIZONA STATE COUNTY

Apache	2	1
Cochise	2	3
Coconino.....	2	5
Gila	2	7
Graham	2	9
Greenlee.....	2	11
Maricopa.....	2	13
Mohave	2	15
Navajo.....	2	17
Pima	2	19
Pinal	2	21
Santa Cruz.....	2	23
Yavapai	2	25
Yuma.....	2	27

ARKANSAS STATE COUNTY

Arkansas	3	1
Ashley	3	3
Baxter.....	3	5
Benton.....	3	7
Boone.....	3	9
Bradley.....	3	11
Calhoun.....	3	13
Carroll.....	3	15

Arkansas (Cont.)	State	County
Chicot.....	3	17
Clark.....	3	19
Clay	3	21
Cleburne.....	3	23
Cleveland	3	25
Columbia.....	3	27
Conway	3	29
Craighead.....	3	31
Crawford	3	33
Crittenden.....	3	35
Cross	3	37
Dallas	3	39
Desha	3	41
Drew.....	3	43
Faulkner	3	45
Franklin.....	3	47
Fulton	3	49
Garland.....	3	51
Grant	3	53
Greene.....	3	55
Hempstead.....	3	57
Hot Springs	3	59
Howard.....	3	61
Independence	3	63
Izard	3	65
Jackson.....	3	67
Jefferson.....	3	69
Johnson	3	71
Lafayette	3	73
Lawrence.....	3	75
Lee	3	77
Lincoln	3	79
Little River	3	81
Logan	3	83
Lonoke	3	85
Madison	3	87
Marion.....	3	89
Miller	3	91
Mississippi	3	93
Monroe.....	3	95

Arkansas (Cont.)	State	County
Montgomery	3	97
Nevada	3	99
Newton.....	3	101
Ouachita.....	3	103
Perry.....	3	105
Phillips	3	107
Pike	3	109
Poinsett	3	111
Polk.....	3	113
Pope	3	115
Prairie.....	3	117
Pulaski.....	3	119
Randolph.....	3	121
St Francis	3	123
Saline	3	125
Scott	3	127
Searcy	3	129
Sebastian.....	3	131
Sevier	3	133
Sharp.....	3	135
Stone	3	137
Union	3	139
Van Buren.....	3	141
Washington.....	3	143
White	3	145
Woodruff	3	147
Yell	3	149

CALIFORNIA STATE COUNTY

Alameda.....	4	1
Alpine	4	3
Amador	4	5
Butte.....	4	7
Calaveras.....	4	9
Colusa	4	11
Contra Costa	4	13
Del Norte	4	15
El Dorado.....	4	17
Fresno	4	19
Glenn	4	21

California (Cont.)	State	County
Humboldt	4	23
Imperial.....	4	25
Inyo	4	27
Kern	4	29
Kings.....	4	31
Lake	4	33
Lassen	4	35
Los Angeles	4	37
Madera	4	39
Marin.....	4	41
Mariposa	4	43
Mendocino	4	45
Merced	4	47
Modoc	4	49
Mono.....	4	51
Monterey	4	53
Napa	4	55
Nevada	4	57
Orange.....	4	59
Placer	4	61
Plumas.....	4	63
Riverside	4	65
Sacramento.....	4	67
San Benito.....	4	69
San Bernardino	4	71
San Diego.....	4	73
San Francisco	4	75
San Joaquin	4	77
San Luis Obispo.....	4	79
San Mateo	4	81
Santa Barbara.....	4	83
Santa Clara.....	4	85
Santa Cruz.....	4	87
Shasta	4	89
Sierra.....	4	91
Siskiyou	4	93
Solano	4	95
Sonoma	4	97
Stanislaus	4	99

California (Cont.)	State	County
Sutter.....	4	101
Tehama	4	103
Trinity	4	105
Tulare.....	4	107
Tuolumne.....	4	109
Ventura	4	111
Yolo	4	113
Yuba.....	4	115

COLORADO	STATE	COUNTY
Adams	5	1
Alamosa	5	3
Arapahoe.....	5	5
Archuleta.....	5	7
Baca	5	9
Bent.....	5	11
Boulder	5	13
Chaffee.....	5	15
Cheyenne	5	17
Clear Creek	5	19
Conejos	5	21
Costilla.....	5	23
Crowley.....	5	25
Custer.....	5	27
Delta.....	5	29
Denver	5	31
Dolores.....	5	33
Douglas.....	5	35
Eagle	5	37
Elbert.....	5	39
El Paso	5	41
Fremont.....	5	43
Garfield.....	5	45
Gilpin	5	47
Grand	5	49
Gunnison.....	5	51
Hinsdale.....	5	53
Huerfano	5	55
Jackson.....	5	57
Jefferson.....	5	59

Colorado (Cont.)	State	County
Kiowa.....	5	61
Kit Carson	5	63
Lake	5	65
La Plata	5	67
Larimer.....	5	69

Las Animas	5	71
Lincoln	5	73
Logan	5	75
Mesa.....	5	77
Mineral.....	5	79
Moffat	5	81
Montezuma	5	83
Montrose	5	85
Morgan.....	5	87
Otero	5	89
Ouray	5	91
Park	5	93
Phillips	5	95
Pitkin.....	5	97
Prowers	5	99
Pueblo	5	101
Rio Blanco	5	103
Rio Grande.....	5	105
Routt.....	5	107
Saguache	5	109
San Juan	5	111
San Miguel.....	5	113
Sedgwick.....	5	115
Summit.....	5	117
Teller.....	5	119
Washington	5	121
Weld.....	5	123
Yuma.....	5	125

FLORIDA	STATE	COUNTY
Alachua	9	1
Baker	9	3
Bay	9	5
Bradford.....	9	7

Florida (Cont.)	State	County
Brevard	9	9
Broward	9	11
Calhoun.....	9	13
Charlotte	9	15
Citrus.....	9	17
Clay	9	19
Collier	9	21
Columbia.....	9	23
Dade.....	9	25
De Soto	9	27
Dixie	9	29
Duval	9	31
Escambia.....	9	33
Flagler.....	9	35
Franklin.....	9	37
Gadsden	9	39
Gilchrist	9	41
Glades	9	43
Gulf.....	9	45
Hamilton	9	47
Hardee.....	9	49
Hendry	9	51
Hernando	9	53
Highlands.....	9	55
Hillsborough	9	57
Holmes.....	9	59
Indian River	9	61
Jackson.....	9	63
Jefferson.....	9	65
Lafayette	9	67
Lake	9	69
Lee	9	71
Leon	9	73
Levy	9	75
Liberty.....	9	77
Madison	9	79
Manatee.....	9	81
Marion.....	9	83
Martin	9	85

Florida (Cont.)	State	County
Monroe.....	9	87
Nassau.....	9	89
Okaloosa	9	91
Okeechobee.....	9	93
Orange.....	9	95
Osceola.....	9	97
Palm Beach	9	99
Pasco	9	101
Pinellas.....	9	103
Polk	9	105
Putnam	9	107
St Johns	9	109
St Lucie	9	111
Santa Rosa.....	9	113
Sarasota.....	9	115
Seminole	9	117
Sumter	9	119
Suwannee	9	121
Taylor.....	9	123
Union	9	125
Volusia	9	127
Wakulla.....	9	129
Walton.....	9	131
Washington	9	133

ILLINOIS	STATE	COUNTY
Adams	12	1
Alexander.....	12	3
Bond.....	12	5
Boone	12	7
Brown.....	12	9
Bureau	12	11
Calhoun.....	12	13
Carroll	12	15
Cass.....	12	17
Champaign	12	19
Christian.....	12	21
Clark.....	12	23
Clay	12	25
Clinton	12	27

Illinois (Cont.)	State	County
Coles	12	29
Cook.....	12	31
Crawford.....	12	33
Cumberland.....	12	35
De Kalb.....	12	37
De Witt.....	12	39
Douglas.....	12	41
Du Page.....	12	43
Edgar.....	12	45
Edwards	12	47
Effingham	12	49
Fayette.....	12	51
Ford.....	12	53
Franklin.....	12	55
Fulton.....	12	57
Gallatin	12	59
Greene.....	12	61
Grundy	12	63
Hamilton	12	65
Hancock	12	67
Hardin	12	69
Henderson.....	12	71
Henry	12	73
Iroquois.....	12	75
Jackson.....	12	77
Jasper	12	79
Jefferson.....	12	81
Jersey	12	83
Jo Daviess	12	85
Johnson	12	87
Kane	12	89
Kankakee	12	91
Kendall.....	12	93
Knox	12	95
Lake	12	97
La Salle	12	99
Lawrence.....	12	101
Lee	12	103
Livingston.....	12	105
Logan.....	12	107

Illinois (Cont.)	State	County
Mc Donough	12	109
Mc Henry	12	111
Mc Lean	12	113
Macon	12	115
Macoupin	12	117
Madison	12	119
Marion.....	12	121
Marshall	12	123
Mason.....	12	125
Massac	12	127
Menard	12	129
Mercer.....	12	131
Monroe.....	12	133
Montgomery.....	12	135
Morgan.....	12	137
Moultrie	12	139
Ogle.....	12	141
Peoria	12	143
Perry.....	12	145
Piatt	12	147
Pike	12	149
Pope	12	151
Pulaski.....	12	153
Putnam	12	155
Randolph.....	12	157
Richland.....	12	159
Rock Island	12	161
St Clair	12	163
Saline	12	165
Sangamon.....	12	167
Schuyler	12	169
Scott	12	171
Shelby	12	173
Stark	12	175
Stephenson.....	12	177
Tazewell.....	12	179
Union	12	181
Vermilion	12	183
Wabash.....	12	185

Illinois (Cont.)	State	County
Warren	12	187
Washington	12	189
Wayne	12	191
White	12	193
Whiteside	12	195
Will	12	197
Williamson	12	199
Winnebago	12	201
Woodford	12	203

INDIANA	STATE	COUNTY
Adams	13	1
Allen	13	3
Bartholomew	13	5
Benton	13	7
Blackford	13	9
Boone	13	11
Brown	13	13
Carroll	13	15
Cass	13	17
Clark	13	19
Clay	13	21
Clinton	13	23
Crawford	13	25
Daviess	13	27
Dearborn	13	29
Decatur	13	31
De Kalb	13	33
Delaware	13	35
Dubois	13	37
Elkhart	13	39
Fayette	13	41
Floyd	13	43
Fountain	13	45
Franklin	13	47
Fulton	13	49
Gibson	13	51
Grant	13	53
Greene	13	55

Indiana (Cont.)	State	County
Hamilton	13	57
Hancock	13	59
Harrison	13	61
Hendricks	13	63
Henry	13	65
Howard	13	67
Huntington	13	69
Jackson	13	71
Jasper	13	73
Jay	13	75
Jefferson	13	77
Jennings	13	79
Johnson	13	81
Knox	13	83
Kosciusko	13	85
Lagrange	13	87
Lake	13	89
La Porte	13	91
Lawrence	13	93
Madison	13	95
Marion	13	97
Marshall	13	99
Martin	13	101
Miami	13	103
Monroe	13	105
Montgomery	13	107
Morgan	13	109
Newton	13	111
Noble	13	113
Ohio	13	115
Orange	13	117
Owen	13	119
Parke	13	121
Perry	13	123
Pike	13	125
Porter	13	127
Posey	13	129
Pulaski	13	131
Putnam	13	133

Indiana (Cont.)	State	County
Randolph.....	13	135
Ripley.....	13	137
Rush.....	13	139
St Joseph.....	13	141
Scott.....	13	143
Shelby.....	13	145
Spencer.....	13	147
Starke.....	13	149
Steuben.....	13	151
Sullivan.....	13	153
Switzerland.....	13	155
Tippecanoe.....	13	157
Tipton.....	13	159
Union.....	13	161
Vanderburgh.....	13	163
Vermillion.....	13	165
Vigo.....	13	167
Wabash.....	13	169
Warren.....	13	171
Warrick.....	13	173
Washington.....	13	175
Wayne.....	13	177
Wells.....	13	179
White.....	13	181
Whitley.....	13	183

KANSAS	STATE	COUNTY
Allen.....	15	1
Anderson.....	15	3
Atchison.....	15	5
Barber.....	15	7
Barton.....	15	9
Bourbon.....	15	11
Brown.....	15	13
Butler.....	15	15
Chase.....	15	17
Chautauqua.....	15	19
Cherokee.....	15	21
Cheyenne.....	15	23

Kansas (Cont.)	State	County
Clark.....	15	25
Clay.....	15	27
Cloud.....	15	29
Coffey.....	15	31
Comanche.....	15	33
Cowley.....	15	35
Crawford.....	15	37
Decatur.....	15	39
Dickinson.....	15	41
Doniphan.....	15	43
Douglas.....	15	45
Edwards.....	15	47
Elk.....	15	49
Ellis.....	15	51
Ellsworth.....	15	53
Finney.....	15	55
Ford.....	15	57
Franklin.....	15	59
Geary.....	15	61
Gove.....	15	63
Graham.....	15	65
Grant.....	15	67
Gray.....	15	69
Greeley.....	15	71
Greenwood.....	15	73
Hamilton.....	15	75
Harper.....	15	77
Harvey.....	15	79
Haskell.....	15	81
Hodgeman.....	15	83
Jackson.....	15	85
Jefferson.....	15	87
Jewell.....	15	89
Johnson.....	15	91
Kearny.....	15	93
Kingman.....	15	95
Kiowa.....	15	97
Labette.....	15	99

Kansas (Cont.)	State	County
Lane	15	101
Leavenworth	15	103
Lincoln	15	105
Linn	15	107
Logan	15	109
Lyon	15	111
Mc Pherson	15	113
Marion	15	115
Marshall	15	117
Meade	15	119
Miami	15	121
Mitchell	15	123
Montgomery	15	125
Morris	15	127
Morton	15	129
Nemaha	15	131
Neosho	15	133
Ness	15	135
Norton	15	137
Osage	15	139
Osborne	15	141
Ottawa	15	143
Pawnee	15	145
Phillips	15	147
Pottawatomie	15	149
Pratt	15	151
Rawlins	15	153
Reno	15	155
Republic	15	157
Rice	15	159
Riley	15	161
Rooks	15	163
Rush	15	165
Russell	15	167
Saline	15	169
Scott	15	171
Sedgwick	15	173
Seward	15	175
Shawnee	15	177
Sheridan	15	179

Kansas (Cont.)	State	County
Sherman	15	181
Smith	15	183
Stafford	15	185
Stanton	15	187
Stevens	15	189
Sumner	15	191
Thomas	15	193
Trego	15	195
Wabaunsee	15	197
Wallace	15	199
Washington	15	201
Wichita	15	203
Wilson	15	205
Woodson	15	207
Wyandotte	15	209

KENTUCKY	STATE	COUNTY
Adair	16	1
Allen	16	3
Anderson	16	5
Ballard	16	7
Barren	16	9
Bath	16	11
Bell	16	13
Boone	16	15
Bourbon	16	17
Boyd	16	19
Boyle	16	21
Bracken	16	23
Breathitt	16	25
Breckinridge	16	27
Bullitt	16	29
Butler	16	31
Caldwell	16	33
Calloway	16	35
Campbell	16	37
Carlisle	16	39
Carroll	16	41
Carter	16	43
Casey	16	45

Kentucky (Cont.)	State	County
Christian.....	16	47
Clark	16	49
Clay.....	16	51
Clinton	16	53
Crittenden	16	55
Cumberland.....	16	57
Daviess.....	16	59
Edmonson	16	61
Elliott	16	63
Estill.....	16	65
Fayette.....	16	67
Fleming	16	69
Floyd.....	16	71
Franklin.....	16	73
Fulton.....	16	75
Gallatin	16	77
Garrard.....	16	79
Grant	16	81
Graves.....	16	83
Grayson.....	16	85
Green	16	87
Greenup	16	89
Hancock.....	16	91
Hardin	16	93
Harlan	16	95
Harrison	16	97
Hart	16	99
Henderson.....	16	101
Henry	16	103
Hickman.....	16	105
Hopkins.....	16	107
Jackson.....	16	109
Jefferson.....	16	111
Jessamine	16	113
Johnson	16	115
Kenton	16	117
Knott	16	119
Knox	16	121

Kentucky (Cont.)	State	County
Larue	16	123
Laurel	16	125
Lawrence.....	16	127
Lee	16	129
Leslie.....	16	131
Letcher	16	133
Lewis.....	16	135
Lincoln	16	137
Livingston	16	139
Logan	16	141
Lyon	16	143
Mc Cracken.....	16	145
Mc Creary	16	147
Mc Lean	16	149
Madison	16	151
Magoffin	16	153
Marion.....	16	155
Marshall	16	157
Martin.....	16	159
Mason.....	16	161
Meade.....	16	163
Menifee	16	165
Mercer.....	16	167
Metcalfe	16	169
Monroe.....	16	171
Montgomery.....	16	173
Morgan.....	16	175
Muhlenberg.....	16	177
Nelson	16	179
Nicholas	16	181
Ohio	16	183
Oldham.....	16	185
Owen.....	16	187
Owsley	16	189
Pendleton	16	191
Perry.....	16	193
Pike	16	195
Powell	16	197
Pulaski.....	16	199

Kentucky (Cont.)	State	County
Robertson.....	16	201
Rockcastle.....	16	203
Rowan.....	16	205
Russell.....	16	207
Scott.....	16	209
Shelby.....	16	211
Simpson.....	16	213
Spencer.....	16	215
Taylor.....	16	217
Todd.....	16	219
Trigg.....	16	221
Trimble.....	16	223
Union.....	16	225
Warren.....	16	227
Washington.....	16	229
Wayne.....	16	231
Webster.....	16	233
Whitley.....	16	235
Wolfe.....	16	237
Woodford.....	16	239

LOUISIANA	STATE	COUNTY
Acadia.....	17	1
Allen.....	17	3
Ascension.....	17	5
Assumption.....	17	7
Avoyelles.....	17	9
Beauregard.....	17	11
Bienville.....	17	13
Bossier.....	17	15
Caddo.....	17	17
Calcasieu.....	17	19
Caldwell.....	17	21
Cameron.....	17	23
Catahoula.....	17	25
Claiborne.....	17	27
Concordia.....	17	29
De Soto.....	17	31
East Baton Rouge.....	17	33
East Carroll.....	17	35

Louisiana (Cont.)	State	County
East Feliciana.....	17	37
Evangeline.....	17	39
Franklin.....	17	41
Grant.....	17	43
Iberia.....	17	45
Iberville.....	17	47
Jackson.....	17	49
Jefferson.....	17	51
Jefferson Davis.....	17	53
Lafayette.....	17	55
Lafourche.....	17	57
La Salle.....	17	59
Lincoln.....	17	61
Livingston.....	17	63
Madison.....	17	65
Morehouse.....	17	67
Natchitoches.....	17	69
Orleans.....	17	71
Ouachita.....	17	73
Plaquemines.....	17	75
Pointe Coupee.....	17	77
Rapides.....	17	79
Red River.....	17	81
Richland.....	17	83
Sabine.....	17	85
St Bernard.....	17	87
St Charles.....	17	89
St Helena.....	17	91
St James.....	17	93
St John The Baptist.....	17	95
St Landry.....	17	97
St Martin.....	17	99
St Mary.....	17	101
St Tammany.....	17	103
Tangipahoa.....	17	105
Tensas.....	17	107
Terrebonne.....	17	109
Union.....	17	111
Vermilion.....	17	113
Vernon.....	17	115

Louisiana (Cont.)	State	County
Washington	17	117
Webster	17	119
West Baton Rouge	17	121
West Carroll	17	123
West Feliciana	17	125
Winn	17	127

MICHIGAN STATE COUNTY

Alcona.....	21	1
Alger	21	3
Allegan.....	21	5
Alpena.....	21	7
Antrim.....	21	9
Arenac.....	21	11
Baraga.....	21	13
Barry	21	15
Bay.....	21	17
Benzie	21	19
Berrien	21	21
Branch.....	21	23
Calhoun.....	21	25
Cass.....	21	27
Charlevoix.....	21	29
Cheboygan	21	31
Chippewa	21	33
Clare.....	21	35
Clinton	21	37
Crawford	21	39
Delta.....	21	41
Dickinson.....	21	43
Eaton	21	45
Emmet.....	21	47
Genesee.....	21	49
Gladwin.....	21	51
Gogebic.....	21	53
Grand Traverse	21	55
Gratiot	21	57
Hillsdale.....	21	59
Houghton	21	61
Huron	21	63

Michigan (Cont.)	State	County
Ingham	21	65
Ionia	21	67
Iosco.....	21	69
Iron.....	21	71
Isabella	21	73
Jackson.....	21	75
Kalamazoo	21	77
Kalkaska.....	21	79
Kent.....	21	81
Keweenaw	21	83
Lake	21	85
Lapeer	21	87
Leelanau.....	21	89
Lenawee	21	91
Livingston	21	93
Luce	21	95
Mackinac.....	21	97
Macomb	21	99
Manistee.....	21	101
Marquette	21	103
Mason.....	21	105
Mecosta	21	107
Menominee	21	109
Midland	21	111
Missaukee	21	113
Monroe.....	21	115
Montcalm	21	117
Montmorency	21	119
Muskegon.....	21	121
Newaygo	21	123
Oakland.....	21	125
Oceana	21	127
Ogemaw	21	129
Ontonagon.....	21	131
Osceola.....	21	133
Oscoda	21	135
Otsego	21	137
Ottawa.....	21	139
Presque Isle	21	141

Michigan (Cont.)	State	County
Roscommon	21	143
Saginaw.....	21	145
St Clair	21	147
St Joseph	21	149
Sanilac.....	21	151
Schoolcraft.....	21	153
Shiawassee.....	21	155
Tuscola.....	21	157
Van Buren.....	21	159
Washtenaw.....	21	161
Wayne.....	21	163
Wexford.....	21	165

MISSISSIPPI	STATE	COUNTY
Adams.....	23	1
Alcorn	23	3
Amite	23	5
Attala.....	23	7
Benton.....	23	9
Bolivar	23	11
Calhoun.....	23	13
Carroll.....	23	15
Chickasaw	23	17
Choctaw	23	19
Claiborne.....	23	21
Clarke.....	23	23
Clay.....	23	25
Coahoma	23	27
Copiah.....	23	29
Covington	23	31
De Soto	23	33
Forrest.....	23	35
Franklin.....	23	37
George	23	39
Greene.....	23	41
Grenada.....	23	43
Hancock.....	23	45
Harrison	23	47
Hinds.....	23	49

Mississippi (Cont.)	State	County
Holmes	23	51
Humphreys.....	23	53
Issaquena.....	23	55
Itawamba.....	23	57
Jackson.....	23	59
Jasper	23	61
Jefferson.....	23	63
Jefferson Davis	23	65
Jones.....	23	67
Kemper.....	23	69
Lafayette	23	71
Lamar	23	73
Lauderdale	23	75
Lawrence.....	23	77
Leake.....	23	79
Lee	23	81
Leflore.....	23	83
Lincoln	23	85
Lowndes.....	23	87
Madison	23	89
Marion.....	23	91
Marshall	23	93
Monroe.....	23	95
Montgomery.....	23	97
Neshoba	23	99
Newton.....	23	101
Noxubee	23	103
Oktibbeha.....	23	105
Panola.....	23	107
Pearl River	23	109
Perry.....	23	111
Pike	23	113
Pontotoc	23	115
Prentiss.....	23	117
Quitman	23	119
Rankin.....	23	121
Scott	23	123
Sharkey	23	125
Simpson	23	127

Mississippi (Cont.)	State	County
Smith.....	23	129
Stone	23	131
Sunflower.....	23	133
Tallahatchie.....	23	135
Tate	23	137
Tippah.....	23	139
Tishomingo	23	141
Tunica	23	143
Union	23	145
Walthall.....	23	147
Warren	23	149
Washington	23	151
Wayne	23	153
Webster	23	155
Wilkinson.....	23	157
Winston.....	23	159
Yalobusha	23	161
Yazoo.....	23	163

MISSOURI	STATE	COUNTY
Adair	24	1
Andrew	24	3
Atchison.....	24	5
Audrain	24	7
Barry	24	9
Barton	24	11
Bates	24	13
Benton.....	24	15
Bollinger	24	17
Boone.....	24	19
Buchanan	24	21
Butler	24	23
Caldwell.....	24	25
Callaway	24	27
Camden	24	29
Cape Girardeau	24	31
Carroll	24	33
Carter	24	35
Cass.....	24	37

Missouri (Cont.)	State	County
Cedar	24	39
Chariton	24	41
Christian.....	24	43
Clark.....	24	45
Clay	24	47
Clinton	24	49
Cole.....	24	51
Cooper.....	24	53
Crawford	24	55
Dade	24	57
Dallas	24	59
Daviess.....	24	61
De Kalb	24	63
Dent.....	24	65
Douglas	24	67
Dunklin	24	69
Franklin.....	24	71
Gasconade	24	73
Gentry	24	75
Greene.....	24	77
Grundy	24	79
Harrison	24	81
Henry	24	83
Hickory	24	85
Holt	24	87
Howard.....	24	89
Howell.....	24	91
Iron.....	24	93
Jackson.....	24	95
Jasper	24	97
Jefferson.....	24	99
Johnson	24	101
Knox	24	103
Laclede.....	24	105
Lafayette	24	107
Lawrence.....	24	109
Lewis.....	24	111
Lincoln	24	113

Missouri (Cont.)	State	County
Linn.....	24	115
Livingston.....	24	117
Mc Donald.....	24	119
Macon.....	24	121
Madison.....	24	123
Maries.....	24	125
Marion.....	24	127
Mercer.....	24	129
Miller.....	24	131
Mississippi.....	24	133
Moniteau.....	24	135
Monroe.....	24	137
Montgomery.....	24	139
Morgan.....	24	141
New Madrid.....	24	143
Newton.....	24	145
Nodaway.....	24	147
Oregon.....	24	149
Osage.....	24	151
Ozark.....	24	153
Pemiscot.....	24	155
Perry.....	24	157
Pettis.....	24	159
Phelps.....	24	161
Pike.....	24	163
Platte.....	24	165
Polk.....	24	167
Pulaski.....	24	169
Putnam.....	24	171
Ralls.....	24	173
Randolph.....	24	175
Ray.....	24	177
Reynolds.....	24	179
Ripley.....	24	181
St Charles.....	24	183
St Clair.....	24	185
St Francois.....	24	187
St Louis.....	24	189

Missouri (Cont.)	State	County
St Louis City.....	24	191
Ste Genevieve.....	24	193
Saline.....	24	195
Schuyler.....	24	197
Scotland.....	24	199
Scott.....	24	201
Shannon.....	24	203
Shelby.....	24	205
Stoddard.....	24	207
Stone.....	24	209
Sullivan.....	24	211
Taney.....	24	213
Texas.....	24	215
Vernon.....	24	217
Warren.....	24	219
Washington.....	24	221
Wayne.....	24	223
Webster.....	24	225
Worth.....	24	227
Wright.....	24	229

MONTANA	STATE	COUNTY
Beaverhead.....	25	1
Big Horn.....	25	3
Blaine.....	25	5
Broadwater.....	25	7
Carbon.....	25	9
Carter.....	25	11
Cascade.....	25	13
Chouteau.....	25	15
Custer.....	25	17
Daniels.....	25	19
Dawson.....	25	21
Deer Lodge.....	25	23
Fallon.....	25	25
Fergus.....	25	27
Flathead.....	25	29
Gallatin.....	25	31
Garfield.....	25	33
Glacier.....	25	35

Montana (Cont.)	State	County
Golden Valley	25	37
Granite	25	39
Hill	25	41
Jefferson.....	25	43
Judith Basin	25	45
Lake	25	47
Lewis And Clark.....	25	49
Liberty.....	25	51
Lincoln.....	25	53
Mc Cone.....	25	55
Madison	25	57
Meagher	25	59
Mineral.....	25	61
Missoula.....	25	63
Musselshell	25	65
Park.....	25	67
Petroleum.....	25	69
Phillips	25	71
Pondera	25	73
Powder River	25	75
Powell	25	77
Prairie.....	25	79
Ravalli.....	25	81
Richland.....	25	83
Roosevelt	25	85
Rosebud	25	87
Sanders.....	25	89
Sheridan	25	91
Silver Bow	25	93
Stillwater.....	25	95
Sweet Grass	25	97
Teton.....	25	99
Toole	25	101
Treasure	25	103
Valley.....	25	105
Wheatland	25	107
Wibaux.....	25	109
Yellowstone	25	111

Montana (Cont.)	State	County
Yellowstone Natl Park	25	113
NEBRASKA	STATE	COUNTY
Adams	26	1
Antelope.....	26	3
Arthur.....	26	5
Banner.....	26	7
Blaine	26	9
Boone	26	11
Box Butte	26	13
Boyd.....	26	15
Brown.....	26	17
Buffalo	26	19
Burt	26	21
Butler	26	23
Cass.....	26	25
Cedar.....	26	27
Chase.....	26	29
Cherry	26	31
Cheyenne	26	33
Clay	26	35
Colfax.....	26	37
Cuming.....	26	39
Custer	26	41
Dakota.....	26	43
Dawes.....	26	45
Dawson	26	47
Deuel.....	26	49
Dixon	26	51
Dodge.....	26	53
Douglas	26	55
Dundy.....	26	57
Fillmore.....	26	59
Franklin	26	61
Frontier.....	26	63
Furnas.....	26	65
Gage	26	67
Garden.....	26	69
Garfield	26	71

Nebraska (Cont.)	State	County
Gosper.....	26	73
Grant	26	75
Greeley.....	26	77
Hall	26	79
Hamilton	26	81
Harlan	26	83
Hayes	26	85
Hitchcock.....	26	87
Holt	26	89
Hooker	26	91
Howard	26	93
Jefferson.....	26	95
Johnson	26	97
Kearney.....	26	99
Keith	26	101
Keya Paha	26	103
Kimball	26	105
Knox	26	107
Lancaster.....	26	109
Lincoln.....	26	111
Logan	26	113
Loup.....	26	115
Mc Pherson	26	117
Madison	26	119
Merrick	26	121
Morrill.....	26	123
Nance	26	125
Nemaha	26	127
Nuckolls.....	26	129
Otoe	26	131
Pawnee	26	133
Perkins	26	135
Phelps.....	26	137
Pierce	26	139
Platte	26	141
Polk.....	26	143
Red Willow	26	145
Richardson	26	147
Rock.....	26	149

Nebraska (Cont.)	State	County
Saline	26	151
Sarpy	26	153
Saunders.....	26	155
Scotts Bluff	26	157
Seward	26	159
Sheridan	26	161
Sherman	26	163
Sioux	26	165
Stanton	26	167
Thayer	26	169
Thomas.....	26	171
Thurston.....	26	173
Valley.....	26	175
Washington	26	177
Wayne	26	179
Webster	26	181
Wheeler	26	183
York	26	185

NEVADA	STATE	COUNTY
Churchill	27	1
Clark.....	27	3
Douglas	27	5
Elko.....	27	7
Esmeralda.....	27	9
Eureka	27	11
Humboldt	27	13
Lander	27	15
Lincoln	27	17
Lyon.....	27	19
Mineral.....	27	21
Nye.....	27	23
Ormsby.....	27	25
Pershing	27	27
Storey	27	29
Washoe.....	27	31
White Pine.....	27	33

NEW MEXICO	STATE	COUNTY
Bernalillo	30	1

New Mexico (Cont.)	State	County
Catron	30	3
Chaves.....	30	5
Colfax	30	7
Curry	30	9
De Baca.....	30	11
Dona Ana.....	30	13
Eddy	30	15
Grant	30	17
Guadalupe	30	19
Harding	30	21
Hidalgo	30	23
Lea	30	25
Lincoln.....	30	27
Luna	30	29
McKinley	30	31
Mora.....	30	33
Otero	30	35
Quay.....	30	37
Rio Arriba	30	39
Roosevelt	30	41
Sandoval	30	43
San Juan	30	45
San Miguel.....	30	47
Santa Fe	30	49
Sierra.....	30	51
Socorro.....	30	53
Taos	30	55
Torrance.....	30	57
Union	30	59
Valencia	30	61

NEW YORK	STATE	COUNTY
Albany.....	31	1
Allegany.....	31	3
Bronx	31	5
Broome	31	7
Cattaraugus	31	9
Cayuga	31	11
Chautauqua	31	13

New York (Cont.)	State	County
Chemung	31	15
Chenango	31	17
Clinton	31	19
Columbia.....	31	21
Cortland	31	23
Delaware	31	25
Dutchess.....	31	27
Erie.....	31	29
Essex	31	31
Franklin.....	31	33
Fulton	31	35
Genesee	31	37
Greene.....	31	39
Hamilton	31	41
Herkimer	31	43
Jefferson.....	31	45
Kings.....	31	47
Lewis.....	31	49
Livingston	31	51
Madison	31	53
Monroe.....	31	55
Montgomery.....	31	57
Nassau	31	59
New York.....	31	61
Niagara.....	31	63
Oneida	31	65
Onondaga	31	67
Ontario	31	69
Orange.....	31	71
Orleans	31	73
Oswego	31	75
Otsego	31	77
Putnam	31	79
Queens	31	81
Rensselaer	31	83
Richmond.....	31	85
Rockland	31	87
St Lawrence	31	89

New York (Cont.)	State	County
Saratoga	31	91
Schenectady	31	93
Schoharie	31	95
Schuyler	31	97
Seneca	31	99
Steuben	31	101
Suffolk	31	103
Sullivan	31	105
Tioga	31	107
Tompkins	31	109
Ulster.....	31	111
Warren	31	113
Washington	31	115
Wayne	31	117
Westchester	31	119
Wyoming	31	121
Yates	31	123

NORTH

DAKOTA	STATE	COUNTY
Adams	33	1
Barnes	33	3
Benson	33	5
Billings.....	33	7
Bottineau.....	33	9
Bowman.....	33	11
Burke	33	13
Burleigh	33	15
Cass.....	33	17
Cavalier.....	33	19
Dickey.....	33	21
Divide	33	23
Dunn	33	25
Eddy	33	27
Emmons	33	29
Foster	33	31
Golden Valley	33	33
Grand Forks	33	35
Grant	33	37
Griggs	33	39

North Dakota (Cont.)	State	County
Hettinger	33	41
Kidder	33	43
La Moure.....	33	45
Logan	33	47
Mc Henry	33	49
Mc Intosh	33	51
Mc Kenzie	33	53
McLean	33	55
Mercer	33	57
Morton	33	59
Mountrail	33	61
Nelson	33	63
Oliver	33	65
Pembina	33	67
Pierce	33	69
Ramsey.....	33	71
Ransom	33	73
Renville	33	75
Richland	33	77
Rolette	33	79
Sargent	33	81
Sheridan	33	83
Sioux	33	85
Slope	33	87
Stark	33	89
Steele.....	33	91
Stutsman.....	33	93
Towner	33	95
Traill.....	33	97
Walsh	33	99
Ward	33	101
Wells	33	103
Williams.....	33	105

OHIO	STATE	COUNTY
Adams	34	1
Allen.....	34	3
Ashland	34	5
Ashtabula	34	7
Athens	34	9

Ohio (Cont.)	State	County
Auglaize.....	34	11
Belmont.....	34	13
Brown	34	15
Butler	34	17
Carroll.....	34	19
Champaign.....	34	21
Clark	34	23
Clermont	34	25
Clinton	34	27
Columbiana.....	34	29
Coshocton	34	31
Crawford.....	34	33
Cuyahoga	34	35
Darke	34	37
Defiance.....	34	39
Delaware	34	41
Erie.....	34	43
Fairfield.....	34	45
Fayette.....	34	47
Franklin.....	34	49
Fulton.....	34	51
Gallia.....	34	53
Geauga	34	55
Greene.....	34	57
Guernsey	34	59
Hamilton	34	61
Hancock	34	63
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Harrison	34	67
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Sakhalin Island.....	101	37
Tadzhik	101	41
Ukraine	101	45
UNITED KINGDOM	110	
England	110	130
North Sea	110	150
Scotland	110	140
Yugoslavia	121	
NORTH AMERICA		
Dominion Of Canada		
Alberta	61	
Medicine Hat	61	1

Lethbridge	61	2
Macleod	61	3
Acadia	61	4
Bow River	61	5
Calgary	61	6
Battle River	61	7
Red Deer	61	8
Camrose	61	10
Wetaskiwin	61	11
Vegreville.....	61	12
Athabaska.....	61	13
Peace River	61	15
Jasper Edson	61	17
British Columbia.....	62	
Division 10 (Ft. St. John).....	62	10
Manitoba.....	63	
Souris	63	4
Brandon.....	63	7
New Brunswick.....	64	
Albert	64	8
Newfoundland	65	
Division 4 (St. Georges-		
Port Au Port).....	65	4
Northwest Territories	66	
Mckenzie (Ft. Norman		
Area)	66	3
Ontario	68	
Brant.....	68	2
Elgin.....	68	9
Haldimand.....	68	15
Kent.....	68	20
Lambton	68	21
Manitoulin.....	68	26
Middlesex.....	68	27
Norfolk.....	68	30
Simcoe	68	43
Quebec.....	70	
Gaspe	70	20
Saskatchewan	71	

Assiniboia	71	1
Weyburn	71	2
Wood Mountain.....	71	3
Maple Creek.....	71	4
Qu'Appelle	71	5
Swift Current	71	8
Battleford	71	13

Mexico..... 111

Nuevo Leon	111	33
Oaxaca	111	35
Tabasco	111	49
Tamaulipas.....	111	51
Veracruz.....	111	55

OCEANIA

Australia 102

Victoria	102	11
Queensland	102	5
Western	102	13

Indonesia..... 83-85

Borneo (Kalimantan)	83	
Tarakan	83	3
Java	84	
Sumatra	85	

New Zealand..... 109

New Plymouth Area.....	109	1
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Sarawak 122 1

SOUTH AMERICA

Argentina..... 112

Chubut	112	5
Neuquen.....	112	29

Brazil..... 123

Chile 113

Magallanes	113	25
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Colombia..... 114

Antioquia	114	3
Bolivar	114	9
Boyaca	114	11
Magdalena.....	114	27
Santander	114	35
Santander Del Norte.....	114	37

Ecuador..... 115

Peru	116	
Piura.....	116	35

Venezuela..... 117

Delta Amacuro	117	1
Anzoategui	117	5
Guarico.....	117	23
Monagas.....	117	33
Zulia	117	47
Lake Maracaibo	117	25

WEST INDIES

Cuba..... 118

Camaguey	118	1
Havana	118	5

Trinidad 119



Table C-1 Oklahoma Sections and County Codes

Region 1		Region 2		Region 3		Region 4		Region 5		Region 6		Region 7	
Alafla	3	Osage	113	Adair	1	Cleveland	27	Candian	17	Atoka	5	Caddo	15
Beaver	7			Cherokee	21	Grady	51	Logan	83	Bryan	13	Comanche	31
Beckhan	9			Craig	35	Hughes	63	Oklahoma	109	Carter	19	Cotton	33
Blaine	11			Creek	37	Lincoln	81			Choctaw	23	Greer	55
Cimarron	25			Delaware	41	McClain	87			Coal	29	Harmon	57
Custer	39			LeFlore	79	Okfuskee	107			Garvin	49	Jackson	65
Ellis	45			Mayes	97	Pottawatomie	125			Johnston	69	Kiowa	75
Garfield	47			Muskogee	101	Seminole	133			Latimer	77	Stephens	137
Grant	53			Nowata	105	Payne	119			Love	85	Tillman	141
Dewey	43			McIntosh	91					Haskel	61	Jefferson	67
Harper	59			Okmulgee	111					McCurtain	89	Washita	149
Kay	71			Ottawa	115					Marshall	95		
Kingfisher	73			Pawnee	117					Murray	99		
Major	93			Rogers	131					Pittsburg	121		
Noble	103			Sequoyah	135					Pontotoc	123		
Roger Mills	129			Tulsa	143					Pushmataha	127		
Texas	139			Wagoner	145								
Woods	151			Washington	147								
Woodward	153												



Table C-2 Texas Sections and County Codes

Region 1		Region 2		Region 3		Region 4		Region 5		Region 6	
Atascosa	3	Bee	25	Austin	15	Aransas	7	Bosque	35	Anderson	1
Bandera	7	Calhoun	57	Brazoria	39	Brooks	47	Collin	85	Angelina	5
Bastrop	9	Dewitt	123	Brazos	41	Cameron	61	Dallas	113	Bowie	37
Bell	11	Goliad	175	Burleson	51	Duval	131	Delta	119	Camp	63
Bexar	25	Jackson	239	Chambers	71	Hidalgo	215	Ellis	139	Cass	67
Blanco	39	Karnes	255	Colorado	89	Jim Hogg	247	Falls	145	Cherokee	73
Burnet	45	Lavaca	285	Fayette	149	Jim Wells	249	Fannin	147	Franklin	159
Cladwell	47	Live Oak	297	Fort Bend	157	Kenedy	261	Freestone	161	Gregg	183
Comal	53	Refugio	391	Galveston	167	Kleberg	273	Henderson	213	Harrison	203
Dimmit	43	Vistoria	469	Grimes	185	Nueces	355	Hill	217	Houston	225
Edwards	59			Hardin	199	San Patricio	409	Hopkins	223	Marion	315
Frio	71			Harris	201	Starr	427	Hunt	231	Morris	343
Gillespie	73			Jasper	241	Webb	479	Johnson	251	Nacogdoches	347
Gonzales	93			Jefferson	245	Willacy	489	Kaufman	257	Panola	365
Guadalupe	103			Lee	287	Zapata	505	Lamar	277	Red River	387
Hays	129			Liberty	291			Leon	289	Rusk	401
Kendall	139			Madison	313			Limestone	293	Sabine	403
Kerr	151			Matagorda	321			McLennan	309	San Augustine	405
Kinney	153			Montgomery	339			Navarro	349	Shelby	419
LaSalle	283			Newton	351			Rains	379	Smith	423
Llano	299			Orange	361			Robertson	395	Titus	449
McMullen	311			Polk	373			Rockwell	397	Upshur	459
Mason	319			San Jacinto	407			Tarrant	439	Wood	499
Maverick	323			Trinity	455			Van Zandt	467		
Medina	325			Tyler	457						
Milam	331			Walker	471						
Real	385			Waller	473						
Travis	453			Washington	477						
Uvalde	463			Wharton	481						
ValVerde	465										
Williamson	491										
Wilson	493										
Zavala	507										

Texas Sections and County Codes

Region 7		Region 8		Region 9		Region 10		Region 11		Region 12	
Brown	49	Coke	81	Andrews	3	Bailey	17	Archer	9	Armstrong	11
Callahan	59	Concho	95	Brewster	43	Borden	33	Baylor	23	Briscoe	45
Coleman	83	Crockett	105	Crane	103	Cochran	79	Clay	77	Carson	65
Comanche	93	Irion	235	Culberson	109	Cottle	101	Cooke	97	Castro	69
Coryell	99	Kimble	267	Ector	135	Crosby	107	Denton	121	Childress	75
Eastland	133	McCulloch	307	El Paso	141	Dawson	115	Foard	155	Collingsworth	87
Erath	143	Menard	327	Glasscock	173	Dickens	125	Grayson	181	Dallam	111
Fisher	151	Reagan	383	Howard	227	Floyd	153	Hardeman	197	Deaf Smith	117
Hamilton	193	Runnels	399	Hudspeth	167	Gaines	165	Jack	237	Donley	129
Haskell	207	Schleicher	413	Jeff Davis	243	Garza	169	Knox	275	Gray	179
Hood	221	Sutton	435	Loving	301	Hale	189	Montague	337	Hall	191
Jones	253	Terrell	443	Martin	317	Hockley	219	Wichita	485	Hansford	195
Lampasas	281	Tom Green	451	Midland	329	Kent	263	Walbarger	487	Hartley	205
Mills	333	Upton	461	Mitchell	335	King	269	Wise	497	Hemphill	211
Nolan	353			Pecos	371	Lamb	279	Young	503	Hutchinson	233
Palo Pinta	363			Presidio	377	Lubbock	303			Lipscomb	295
Parker	367			Reeves	389	Lynn	305			Moore	341
San Saba	411			Sterling	431	Motely	345			Ochiltree	357
Shackelford	417			Ward	475	Scurry	415			Oldham	359
Somervell	425			Winkler	495	Terry	445			Parmer	369
Stephens	429					Yoakum				Potter	375
Stonewall	433									Randall	381
Taylor	441									Roberts	393
Throckmorton	447									Sherman	421
										Swisher	437
										Wheeler	483

APPENDIX D

Example of an Analysis by the Bureau of Mines Routine Method

BPO CRUDE OIL DATA BANK
MAINTAINED BY NATIONAL INSTITUTE FOR PETROLEUM AND ENERGY RESEARCH (NIPER)
BARTLESVILLE, OK
CRUDE OIL ANALYSIS NO. B68090

IDENTIFICATION

LAKE RACCOURCI FIELD

LOUISIANA
LAFOURCHE

12404 - 13456 FEET

GENERAL CHARACTERISTICS

GRAVITY, SPECIFIC: 0.832 API: 38.6 POUR POINT, DEG F: < 5
SULFUR, WT PCT: 0.190 NITROGEN, WT PCT: 0.043
VISCOSITY, SAYBOLT UNIV SEC: COLOR: GREENISH-BROWN
 AT 77 F 38
 AT 100 F 37

WELL INFORMATION

WELL NUMBER: 4, 4D, 15 LOCATION:
SECTION: TOWNSHIP: RANGE:
LONGITUDE: LATITUDE: COMPLETION DATE:
ADDITIONAL WELL INFORMATION:

DISTILLATION, BUREAU OF MINES ROUTINE METHOD

STAGE 1-DISTILLATION AT ATMOSPHERIC PRESSURE 747 MM HG
FIRST DROP AT 77 DEGREES F.

FRAC CUT NO. TEMP F	VOL PCT	SUM VOL PCT	GRAVITY SPEC (60/60 F)	API	CORR INDEX	REFRACT INDEX, ND (20 C)	SPEC DISP	SU VISC SECS (100 F)	CLOUD POINT F
1	122	1.7	1.7	0.633	92.0				
2	167	2.3	4.0	0.660	82.9	2	1.37372	131.1	
3	212	4.2	8.2	0.702	70.1	12	1.39646	133.0	
4	257	6.4	14.6	0.732	61.8	17	1.40997	132.1	
5	302	5.7	20.3	0.754	56.2	20	1.42099	136.7	
6	347	5.6	25.9	0.772	51.8	22	1.43013	137.9	
7	392	5.2	31.1	0.788	48.1	23	1.43818	138.4	
8	437	6.0	37.1	0.802	44.9	24	1.44565	138.6	
9	482	6.1	43.2	0.817	41.7	26	1.45354	140.8	
10	527	6.2	49.4	0.831	38.8	28	1.46101	145.2	

STAGE 2-DISTILLATION CONTINUED AT 40 MM HG

11	392	7.9	57.3	0.852	34.6	34	1.46972	144.5	43	< 5
12	437	7.7	65.0	0.865	32.1	36	1.47503	147.5	52	30
13	482	6.2	71.2	0.880	29.3	40	1.48269	151.4	73	40
14	527	4.6	75.8	0.890	27.5	42	1.48838		120	60
15	572	4.6	80.4	0.899	25.9	43			255	70
RESIDUUM		18.2	98.6	0.959	16.0					

CARBON RESIDUE, WT PCT: OF RES: 5.6 SULFUR, WT PCT OF RES:
 (CONRADSON) OF CRUDE: 1.2 NITROGEN, WT PCT OF RES:

	APPROXIMATE SUMMARY			
	VOL PCT	SP GRAV	API	VISCOSITY
LIGHT GASOLINE	8.2	0.676	77.8	
TOTAL GASOLINE AND NAPHTHA	31.1	0.738	60.3	
KEROSENE DISTILLATE	12.1	0.810	43.3	
GAS OIL	16.3	0.842	36.5	
NONVISCIOUS LUBE DISTILLATE	11.8	0.862-0.886	32.6-28.3	50-100
MEDIUM LUBE DISTILLATE	5.0	0.886-0.895	28.3-26.5	100-200
VISCOUS LUBE DISTILLATE	4.2	0.895-0.904	26.5-25.1	> 200
RESIDUE	18.2	0.959	16.0	
DISTILLATION LOSS	1.4			

End of Report . . .

Example of a Hydrocarbon-Type Analysis

BPO CRUDE OIL DATA BANK
MAINTAINED BY NATIONAL INSTITUTE FOR PETROLEUM AND ENERGY RESEARCH (NIPER)
BARTLESVILLE, OK

HYDROCARBON-TYPE ANALYSIS CRUDE OIL ANALYSIS NO. B68090

FRAC NO.	VOL PCT OF CRUDE	SPEC GRAV	CORR INDEX	AROMATICS		PAR-NAPH		CORR INDEX OF P-N	SPEC GRAV OF P-N
				VOL PCT OF FRAC		VOL PCT OF FRAC			
1	1.7	0.633	-			100			0.633
2	2.3	0.660	3	5.3		94.7			0.648
3	4.2	0.702	13	7.1		92.9	7		0.689
4	6.4	0.732	18	6.8		93.2	13		0.722
5	5.7	0.754	21	11.1		88.9	14		0.739
6	5.6	0.772	22	12.8		87.2	15		0.756
7	5.2	0.788	24	13.3		86.7	16		0.772
8	6.0	0.802	25	13.3		86.7	17		0.785
9	6.1	0.817	27	14.0		86.0	17		0.797
10	6.2	0.831	29	15.4		84.6	19		0.810
11	7.9	0.852	35	15.5		84.5	26		0.833
12	7.7	0.865	37	17.9		82.1	28		0.845

ANALYSIS OF NAPHTHA FRACTIONS					FRAC NO.	NO. OF RINGS PER MOL:		
FR	VOL PCT	OF PN	VOL PCT	OF FR		TOTAL	AROM	NAPHTHENES
NO.	NAPH	PAR	NAPH	PAR				
2	0.0	100.0	0.0	94.7	12	1.7	0.2	1.5
3	9.3	90.7	8.6	84.3	14	2.3	0.4	1.9
4	32.3	67.7	30.1	63.1				
5	35.3	64.7	31.4	57.5				
6	35.4	64.6	30.9	56.3				
7	43.0	57.0	37.3	49.4				

SUMMARY DATA FOR BLENDS

	NAPHTHA BLEND (FR1-7)	GAS OIL BLEND (FR 8-12)
VOLUME PERCENT OF CRUDE IN BLEND	31.1	33.9
AROMATIC, VOL PCT OF BLEND	9.3	15.4
PARAFFIN-NAPHTHENE, VOL PCT OF BLEND	90.7	84.6
NAPHTHENE, VOL PCT OF BLEND	25.3	
PARAFFIN, VOL PCT OF BLEND	65.4	
NAPHTHENE, VOL PCT OF P-N IN BLEND	27.9	
PARAFFIN, VOL PCT OF P-N IN BLEND	72.1	
NAPHTHENE RING, WT PCT OF P-N IN BLEND	14.4	32.7
PAR + SIDE CHAINS, WT PCT OF P-N IN BLEND	85.6	67.3

End of Report . . .